TOPIC 1.1
Scalars and Vectors

Required Course Content

LEARNING OBJECTIVE	ESSENTIAL KNOWLEDGE
1.1.A Describe a scalar or vector quantity using magnitude and direction, as appropriate.	**1.1.A.1** Scalars are quantities described by magnitude only; vectors are quantities described by both magnitude and direction. **1.1.A.2** Vectors can be visually modeled as arrows with appropriate direction and lengths proportional to their magnitude. **1.1.A.3** Distance and speed are examples of scalar quantities, while position, displacement, velocity, and acceleration are examples of vector quantities. **1.1.A.4** Vectors can be expressed in unit vector notation or as a magnitude and a direction. **1.1.A.4.i** Unit vector notation can be used to represent vectors as the sum of their constituent components in the x-, y-, and z-directions, denoted by $\hat{i}$, $\hat{j}$, and $\hat{k}$, respectively. *Relevant equation:* $\vec{r} = \left(A\hat{i} + B\hat{j} + C\hat{k}\right)$ **1.1.A.4.ii** The position vector of a point is given by $\vec{r}$, and the unit vector in the direction of the position vector is denoted $\hat{r}$.

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SUGGESTED SKILLS

1.A Create diagrams, tables, charts, or schematics to represent physical situations.

2.A Derive a symbolic expression from known quantities by selecting and following a logical mathematical pathway.

2.B Calculate or estimate an unknown quantity with units from known quantities, by selecting and following a logical computational pathway.

3.B Apply an appropriate law, definition, theoretical relationship, or model to make a claim.

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LEARNING OBJECTIVE

1.1.A

Describe a scalar or vector quantity using magnitude and direction, as appropriate.

ESSENTIAL KNOWLEDGE

1.1.A.4.iii

A resultant vector is the vector sum of the addend vectors' components.

Relevant equations:

$$\vec{C} = \vec{A} + \vec{B}$$

$$\vec{C} = (A_x + B_x)\hat{i} + (A_y + B_y)\hat{j}$$

1.1.A.5

In a given one-dimensional coordinate system, opposite directions are denoted by opposite signs.

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